Math Thermo
class #11
02/24/2026

Recap

For N non-interacting particles, with exactly r accessible discrete energy states, the Gibbs entropy,

$$S_G(\vec{\nu}) = -k_B \sum_{j=1}^{r} n_j \log\left(\frac{n_j}{N}\right),$$

is maximized by the Maxwell-Boltzmann distribution,

$$n_j^{eq} = \exp\left(\frac{-U_j}{k_B T}\right) \cdot \frac{N}{Z},$$

where Z is the partition function given by

$$Z = Z(T) = \sum_{i=1}^{r} \exp\left(\frac{-U_i}{k_B T}\right).$$

Recall that the mass and energy constraints hold

$$N = \sum_{j=1}^{r} n_j^{eq}, \qquad U = \sum_{j=1}^{r} n_j^{eq} U_j,$$

where the energy levels U_j, $j=1,\dots,r$, are fixed. All of the Thermodynamic variables are given in terms of Z:

$$F(T,N) = -N k_B T \ln(Z),$$

$$\tilde{U}(T,N) = N k_B T^2 \frac{\partial}{\partial T}\left[\log(Z)\right],$$

$$\tilde{S}_{eq}(T,N) = N k_B \log(Z) + N k_B T \frac{\partial}{\partial T}\left[\log(Z)\right],$$

$$\mu = k_B T \log(Z).$$

Continuous Phase Space

Next we consider the case that the energy levels are continuous.

Again we will only consider the case of N non-interacting particles. The energy of each particle is assumed to be a function of the form

(11.1) $$u(\vec{x}, \vec{v}) = \frac{1}{2} m |\vec{v}|^2 + \psi(\vec{x}),$$

where $\vec{x} \in \Omega$ is the position of the particle in the domain Ω and $\vec{v} \in \mathbb{R}^3$ is its velocity vector.

The energy of any one particle is independent of the position and velocity of the other particles in the system.

m is the mass of the particle.

ψ is a potential energy, which depends upon only the position of the particle.

The term $\frac{1}{2} m |\vec{v}|^2$ is called the kinetic energy.

The expression (11.1) is an idealization because it neglects energetic interactions between particles.

If N is large, it is too hard to track particles individually. We use a density instead.

Definition (11.1): Let $\Omega \subseteq \mathbb{R}^3$ be a simply-connected domain with volume

$$V = \int_{\Omega} d\vec{x}.$$

The number density of particles is a function

$$f : \Omega \times \mathbb{R}^3 \longrightarrow [0, \infty)$$

with the property

(11.2) $$\int_{\Omega} \int_{\mathbb{R}^3} f(\vec{x}, \vec{v}) \, d\vec{v} \, d\vec{x} = N > 0$$

where N is the total number of particles in Ω. The function

$$p : \Omega \times \mathbb{R}^3 \longrightarrow [0, \infty),$$

defined as

$$p(\vec{x}, \vec{v}) := \frac{f(\vec{x}, \vec{v})}{N},$$

is called the particle probability function. Of course

(11.3) $$\int_{\Omega} \int_{\mathbb{R}^3} p(\vec{x}, \vec{v}) \, d\vec{v} \, d\vec{x} = 1.$$

We assume that there is a fixed and finite amount of energy in the system

(11.4) $$U = \int_{\Omega}\int_{\mathbb{R}^3} f(\vec{x},\vec{v})\, u(\vec{x},\vec{v})\, d\vec{v}\, d\vec{x}$$

The set $Q = \Omega \times \mathbb{R}^3$ is called phase space. It is of course in $\mathbb{R}^6$!

Suppose $A \subseteq Q$. The number of particles in A is

$$N \geq N_A := \iint_A f(\vec{x},\vec{v})\, d\vec{v}\, d\vec{x}.$$

The energy of the particles in that set is denoted

$$U \geq U_A := \iint_A f(\vec{x},\vec{v})\, u(\vec{x},\vec{v})\, d\vec{v}\, d\vec{x}.$$

<u>Definition</u> (11.2): The <u>average energy</u> (per particle) is defined as

$$\overline{U} = \iint_Q p(\vec{x},\vec{v})\, u(\vec{x},\vec{v})\, d\vec{v}\, d\vec{x}.$$

Of course, it must be that

(11.5) $$U = N\overline{U}.$$

Note

$$[p] = \frac{s^3}{m^6} = \frac{1}{[\vec{v}]^3[\vec{x}]^3} = \frac{1}{[\text{Vol PhaseSpace}]}.$$

<u>Definition</u> (11.3): The <u>Gibbs entropy</u> for a system of N particles in phase space $Q = \Omega \times \mathbb{R}^3$ with particle density $f: Q \to [0,\infty)$ is defined as

(11.6) $$\mathcal{S}_G[f] := -k_B \iint_Q f(\vec{x},\vec{v}) \log\left(\frac{f(\vec{x},\vec{v})}{P_0 N}\right) d\vec{v}\,d\vec{x},$$

where $P_0 > 0$ is a normalization factor with the property that

$$\left[\frac{f(\vec{x},\vec{v})}{P_0 N}\right] = \left[\frac{p(\vec{x},\vec{v})}{P_0}\right] = 1 \Leftrightarrow [P_0] = \frac{1}{[\vec{v}]^3[\vec{x}]^3}$$

The <u>average entropy</u> (per particle) is defined as

$$\bar{\mathcal{S}}_G[f] = \frac{\mathcal{S}_G[f]}{N}.$$

Thus,

(11.7) $$\bar{\mathcal{S}}_G[f] = -k_B \iint_Q p(\vec{x},\vec{v}) \log\left(\frac{p(\vec{x},\vec{v})}{P_0}\right) d\vec{v}\,d\vec{x}.$$

<u>Remark</u>: The argument for a transcendental function, like log, must be unitless.

As in the discrete (quantized) energy case the equilibrium state is the one that maximizes the Gibbs entropy.

<u>Theorem</u> (11.1): The equilibrium particle density for non-interacting particles with constraints

(11.8)
$$N = \iint_Q f(\vec{x},\vec{v})\, d\vec{v}\, d\vec{x}$$

and

(11.9)
$$U = \iint_Q f(\vec{x},\vec{v})\, u(\vec{x},\vec{v})\, d\vec{v}\, dx$$

is given by the <u>Maxwell - Boltzmann distribution</u>

(11.10)
$$f(\vec{x},\vec{v}) = f^{eq}(\vec{x},\vec{v}) = P_0 \frac{N}{Z} \exp\left(\frac{-\beta\, u(\vec{x},\vec{v})}{k_B}\right),$$

where Z is the partition function

(11.11)
$$Z = P_0 \iint_Q \exp\left(\frac{-\beta\, u(\vec{x},\vec{v})}{k_B}\right) d\vec{v}\, d\vec{x}$$

and β is a Lagrange multiplier that can be uniquely determined as a function of U and N.

<u>Proof</u>: let us minimize the functional

$$J[f] = k_B \iint_Q f(\vec{q}) \log\left(\frac{f(\vec{q})}{P_0 N}\right) d\vec{q}$$

$$+ \alpha\left\{\iint_Q f(\vec{q})\, d\vec{q} - N\right\} + \beta\left\{\iint_Q f(\vec{q})\, u(\vec{q})\, d\vec{q} - U\right\}$$

where

$$\vec{q} := (\vec{x}, \vec{v}) \in Q$$

and $\alpha, \beta \in \mathbb{R}$ are Lagrange multipliers. To compute the variation observe that

$$\frac{d}{ds}\left[J(f+sg)\right]\Big|_{s=0}$$

$$= \iint_Q g(\vec{q})\left\{k_B \log\left(\frac{f(\vec{q})}{NP_0}\right) + k_B + \alpha + \beta\, u(\vec{q})\right\} d\vec{q}$$

$$= 0$$

for any g with $f+sg : Q \to [0,\infty)$, et cetera. Thus,

$$k_B \log\left(\frac{f(\vec{q})}{NP_0}\right) + k_B + \alpha + \beta\, u(\vec{q}) = 0, \qquad \forall\, \vec{q} \in Q.$$

Solving for f, we find

$$f^{eq}(\vec{q}) = P_0 \frac{N}{Z} \exp\left(\frac{-\beta\, u(\vec{q})}{k_B}\right),$$

where

$$Z = P_0 \iint_Q \exp\left(\frac{-\beta\, u(\vec{q})}{k_B}\right) d\vec{q}$$

is the unitless <u>partition function</u>. Here we have used the constraint (11.8) to eliminate α. Thus,

$$\iint_Q f^{eq}(\vec{q})\, d\vec{q} = N.$$

We contend that, as before, the energy constraint (11.9) is sufficient to eliminate β, and in fact β is a unique positive function of U and N.

We will demonstrate this momentarily. ///

<u>Theorem</u> (11.2): At equilibrium, the Gibbs entropy satisfies

(11.12) $$S_{eq}(U,N) := \mathcal{S}_G[f^{eq}] = \beta U + k_B \log(Z)\, N,$$

and at equilibrium the derivatives are

(11.13) $$\frac{\partial S_{eq}}{\partial U} = \beta \quad \text{and} \quad \frac{\partial S_{eq}}{\partial N} = k_B \log(Z).$$

<u>Proof</u>: At equilibrium,

$$f^{eq}(\vec{q}) = P_0 \frac{N}{Z} \exp\left(\frac{-\beta\, u(\vec{q})}{k_B}\right)$$

where

$$Z = P_0 \iint_Q \exp\left(\frac{-\beta\, u(\vec{q})}{k_B}\right) d\vec{q}$$

$$S_{eq}(U,N) = \mathcal{S}_G[f^{eq}]$$
$$= -k_B \iint_Q f^{eq}(\vec{x},\vec{v}) \log\left(\frac{f^{eq}(\vec{x},\vec{v})}{P_0\, N}\right) d\vec{v}\, d\vec{x}$$

$$= -k_B \iint_Q \frac{P_0 N}{Z} \exp\left(\frac{-\beta u(\vec{q})}{k_B}\right) \log\left(\frac{\exp\left(\frac{-\beta u(\vec{q})}{k_B}\right)}{Z}\right) d\vec{q}$$

$$= \frac{-k_B P_0 N}{Z} \iint_Q \exp\left(\frac{-\beta u(\vec{q})}{k_B}\right) \left\{ \frac{-\beta u(\vec{q})}{k_B} - \log(Z) \right\} d\vec{q}$$

$$= \frac{P_0 N}{Z} \beta \iint_Q \exp\left(\frac{-\beta u(\vec{q})}{k_B}\right) u(\vec{q})\, d\vec{q}$$

$$+ \frac{k_B P_0 N}{Z} \log(Z) \iint_Q \exp\left(\frac{-\beta u(\vec{q})}{k_B}\right) d\vec{q}$$

$$= \beta U + k_B N \log(Z).$$

Suppose that τ is a process parameter and assume that $(\vec{q}, \tau) \longmapsto f(\vec{q}, \tau)$,

$$f : Q \times [0, \infty) \longrightarrow [0, \infty).$$

Thus,

$$N(\tau) = \iint_Q f(\vec{q}, \tau)\, d\vec{q}.$$

It now follows that

$$\frac{d}{d\tau} S_G[f(\cdot, \tau)]$$

$$= -k_B \frac{d}{d\tau} \iint_Q f(\vec{q}, \tau) \log\left(\frac{f(\vec{q}, \tau)}{P_0 N(\tau)}\right) d\vec{q}$$

$$= -k_B \iint_Q \left\{ \dot{f} \log\left(\frac{f}{P_0 N}\right) + f\left[\frac{\dot{f}}{f} - \frac{\dot{N}}{N}\right] \right\} d\vec{q}$$

$$= -k_B \iint_Q \dot{f} \left\{ \log\left(\frac{f}{P_0 N}\right) + 1 \right\} d\vec{q}$$

$$+ k_B \frac{\dot{N}}{N} \iint_Q f \, d\vec{q}$$

$$= -k_B \iint_Q \dot{f} \log\left(\frac{f}{P_0 N}\right) d\vec{q}.$$

Here, we have used the notation

$$\dot{f} = \frac{\partial f}{\partial \tau}, \qquad \dot{N} = \frac{dN}{d\tau}.$$

Next, observe that

$$U(\tau) = \iint_Q f(\vec{q}, \tau) \, u(\vec{q}) \, d\vec{q},$$

So that

$$\frac{dU}{d\tau} = \dot{U} = \iint_Q \dot{f} \, u \, d\vec{q}.$$

Suppose that $(\vec{q}, \tau) \longmapsto f(\vec{q}, \tau)$,

$$f : Q \times [0, \infty) \to [0, \infty),$$

is a path through number density space, with

$$f(\cdot,\tau_0) = f^{eq}(\cdot).$$

Then,

$$N(\tau_0) = N, \qquad U(\tau_0) = U,$$

and

$$\frac{d}{d\tau} S_G[f(\cdot,\tau_0)]$$

$$= -k_B \iint_Q \dot{f}(q,\tau_0) \log\left(\frac{f(\vec{q},\tau_0)}{P_0 N(\tau_0)}\right) d\vec{q}$$

$$= -k_B \iint_Q \dot{f}(q,\tau_0) \log\left(\frac{f^{eq}(\vec{q})}{P_0 N}\right) d\vec{q}$$

$$= -k_B \iint_Q \dot{f}\left[\left(-\frac{\beta u(\vec{q})}{k_B}\right) - \log(Z)\right] d\vec{q}$$

$$= \beta \dot{U} + k_B \dot{N} \log(Z).$$

As in the proof of Theorem (10.2), we can conclude that

$$\frac{\partial S_{eq}}{\partial U} = \beta \quad \text{and} \quad \frac{\partial S_{eq}}{\partial N} = k_B \log(Z). \quad ///$$

__Definition (11.3):__ The __statistical temperature__ is defined by the expression

(11.14) $$\frac{1}{T} := \frac{\partial S_{eq}}{\partial U},$$

where

$$S_{eq}(U,N) = S_G[f^{eq}] .$$

The <u>statistical chemical potential</u> is defined via

(11.15) $$\frac{\mu}{T} := \frac{\partial S_{eq}}{\partial N} .$$

Thus

(11.16) $$T = \frac{1}{\beta}$$

and

(11.17) $$\mu = k_B T \log(Z) .$$